Assignment 1 : (Git & GitHub)

Q1. Create and Initialize Git Repository

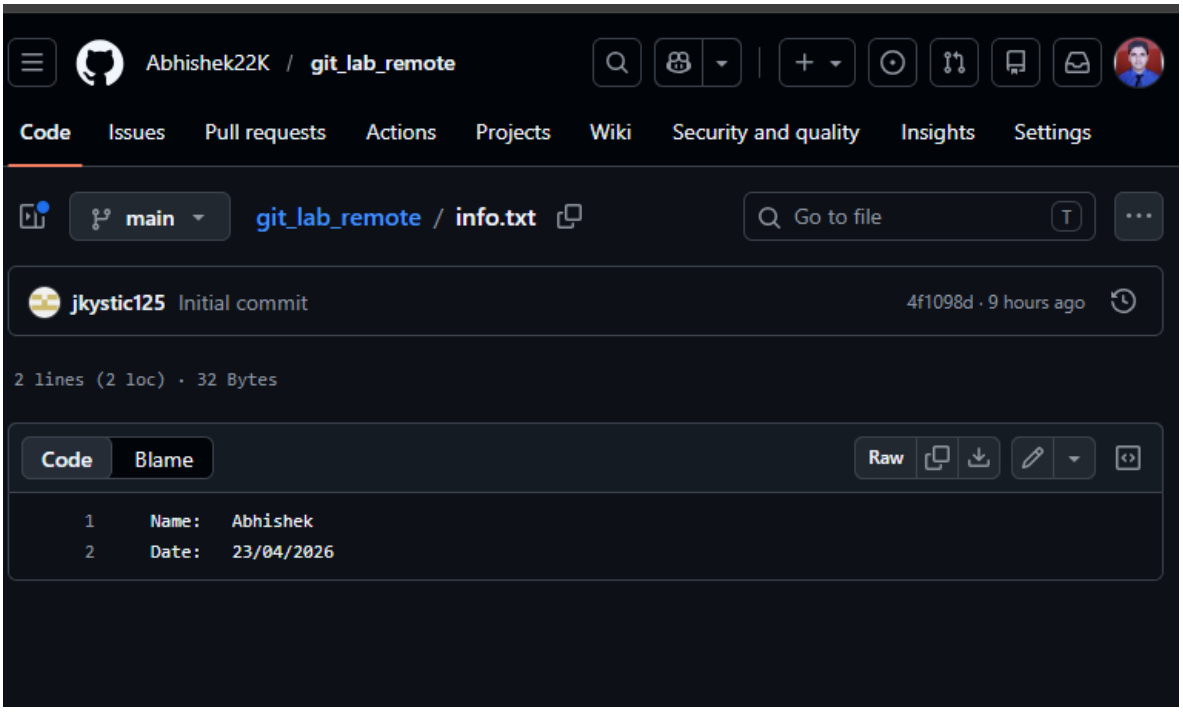
- Create a new repository named `git_lab` on your local system.
- Initialize it using Git.
- Create a file `info.txt` and add your name and date.
- Check the repository status.
- Add the file to staging and commit with message `"Initial commit"`.

```
namo@namo-vmwarevirtualplatform:~$ mkdir git_lab
namo@namo-vmwarevirtualplatform:~$ cd git_lab
namo@namo-vmwarevirtualplatform:~/git_lab$ ls
namo@namo-vmwarevirtualplatform:~/git_lab$ nano info.txt
namo@namo-vmwarevirtualplatform:~/git_lab$ ls
info.txt
namo@namo-vmwarevirtualplatform:~/git_lab$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: is subject to change. To configure the initial branch name to use in all
hint: of your new repositories, which will suppress this warning, call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
Initialized empty Git repository in /home/namo/git_lab/.git/
namo@namo-vmwarevirtualplatform:~/git_lab$ git status
On branch master

No commits yet

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        info.txt

nothing added to commit but untracked files present (use "git add" to track)
namo@namo-vmwarevirtualplatform:~/git_lab$ git add .
namo@namo-vmwarevirtualplatform:~/git_lab$ git commit -m "Initial commit"
[master (root-commit) 4f1098d] Initial commit
 1 file changed, 2 insertions(+)
 create mode 100644 info.txt
```

Q2. Connect to GitHub and Push Code

- Create a repository named `git_lab_remote` on GitHub.
- Link your local repository (`git_lab`) to the remote repository.
- Push your committed file to the remote repository.
- Verify the file is uploaded on GitHub.

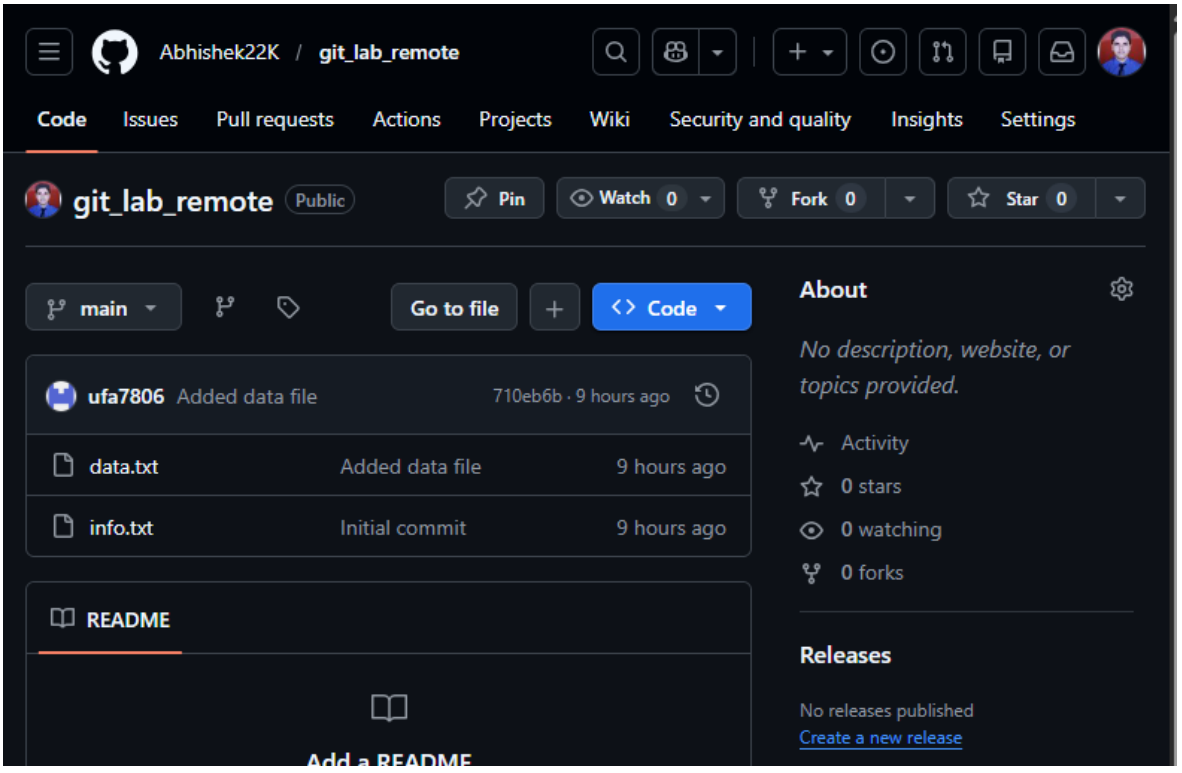
```
namo@namo-vmwarevirtualplatform:~/git_lab$ git push origin main
Username for 'https://github.com': ak
Password for 'https://ak@github.com':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 231 bytes | 231.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Abhishek22K/git_lab_remote.git
 * [new branch]      main -> main
```



Q3. Clone Repository and Make Changes

- Clone the repository `git_lab_remote` to a new directory.
- Create a new file `data.txt` and add some sample content.
- Add and commit the file with message `"Added data file"`.
- Push the changes back to GitHub.

```
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ nano data.txt
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ git init
Reinitialized existing Git repository in /home/namo/clone/git_lab_remote/.git/
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ git add .
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ git commit -m "Added data file"
[main 710eb6b] Added data file
 1 file changed, 1 insertion(+)
 create mode 100644 data.txt
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ git push origin main
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 271 bytes | 271.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0
To https://github.com/Abhishek22K/git_lab_remote.git
   4f1098d..710eb6b  main -> main
```



Q4. Branching and Merging

- Create a new branch named `feature`.
- Switch to the `feature` branch.
- Modify `info.txt` (add new line: “Feature branch update”).
- Commit the changes.
- Switch back to `main` branch.
- Merge the `feature` branch into `main`.
- Delete the `feature` branch after merge.

```
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ git checkout -b feature
Switched to a new branch 'feature'
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ nano info.txt
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ git add info.txt
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ git commit -m "Feature branch update"
[feature adfa460] Feature branch update
 1 file changed, 2 insertions(+)
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ git checkout main
Switched to branch 'main'
Your branch is up to date with 'origin/main'.
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ git merge feature
Updating 710eb6b..adfa460
Fast-forward
 info.txt | 2 ++
 1 file changed, 2 insertions(+)
namo@namo-vmwarevirtualplatform:~/clone/git_lab_remote$ git branch -d feature
Deleted branch feature (was adfa460).
```